

Bayesian Networks II: Sampling and Approximate Inference

Lecture 20

CS 4880

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Last Time: Exact Inference Worked

For the small robot network,

$$P(O, R, C, L) = P(O)P(R)P(C \mid O, R)P(L \mid O, R),$$

and exact inference gave

$$P(O \mid C = +, L = +) \approx 0.902.$$

What we gained

Correctly modeled correlated sensors and obtained an exact posterior.

What it cost

We had to sum over every hidden assignment relevant to the query.

Key idea

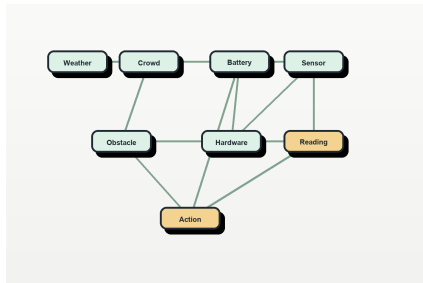
Exact inference is attractive because it is exact—not because it is always cheap.

A Real Robot Network Does Not Stay Small

Add realistic hidden variables:

- ▶ rain, glare, dust, lighting, and surface reflectivity;
- ▶ pedestrian, bicycle, vehicle, and temporary construction states;
- ▶ camera, LiDAR, ultrasonic, wheel-slip, and IMU observations;
- ▶ component health and calibration states.

Each new variable may introduce new CPTs and new hidden assignments.



Key idea

A compact representation can still produce a difficult inference problem.

Exact Inference Must Account for Hidden Variables

Suppose the query is

$$P(O \mid c, l, u, w),$$

where the robot has four observed sensors but many unobserved environmental variables $H_1, \dots, H_k$.
Exact inference requires terms of the form

$$P(O, c, l, u, w) = \sum_{h_1} \cdots \sum_{h_k} P(O, c, l, u, w, h_1, \dots, h_k).$$

The problem

The graph may be compact, but the number of hidden configurations considered during inference can grow very quickly.

Key idea

Representation complexity and inference complexity are related—but they are not the same thing.

The Alternative: Estimate Instead of Enumerate

What if the robot does not need the exact value

$$P(O \mid e) = 0.90231 \dots$$

and would make the same control decision with

$$\hat{P}(O \mid e) \approx 0.90?$$

Exact inference

Deterministically combine factors until the requested probability is computed.

Approximate inference

Generate representative assignments and estimate the requested probability from them.

Key idea

Sampling trades deterministic precision for computational tractability.

A Sample Is One Possible World

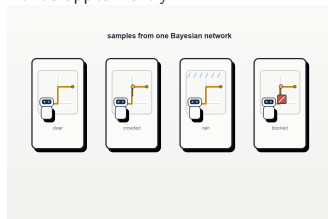
A complete sample might be

$$O = \text{true}, \quad R = \text{false}, \quad C = +, \quad L = +.$$

Another might be

$$O = \text{false}, \quad R = \text{true}, \quad C = +, \quad L = -.$$

If samples are generated according to the Bayesian network, common worlds appear often and unlikely worlds appear rarely.



Key idea

Sampling turns a probability distribution into a stream of concrete simulated assignments.

Monte Carlo Estimation Uses Frequencies

To estimate an event A , generate N samples and count how often A occurs:

$$\hat{P}(A) = \frac{N(A)}{N}.$$

Example: in 10,000 samples, suppose 1,017 contain an obstacle.

$$\hat{P}(O) = \frac{1017}{10000} = 0.1017.$$

The true prior in our model is $P(O) = 0.10$.

Key idea

We replace symbolic summation with empirical counting over simulated worlds.

More Samples Usually Mean Less Monte Carlo Noise

For independent samples, standard Monte Carlo error typically shrinks on the order of

$$\frac{1}{\sqrt{N}}.$$

Samples	Relative error scale
1,000	$1/\sqrt{1000}$
4,000	about half as large
100,000	about one tenth as large as at 1,000

Important tradeoff

Ten times more accuracy does not usually require ten times more samples—it can require roughly one hundred times more.

Key idea

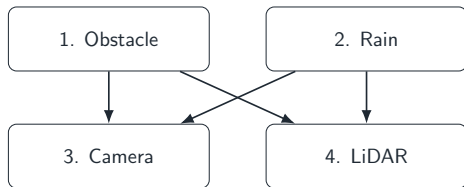
Approximate inference gives us a knob: spend more computation to reduce statistical error.

Prior Sampling Follows the Graph

Generate variables in topological order.

For the robot network:

1. Sample $O \sim P(O)$.
2. Sample $R \sim P(R)$.
3. Sample $C \sim P(C \mid O, R)$.
4. Sample $L \sim P(L \mid O, R)$.



Numbers show one valid topological sampling order; arrows remain the Bayesian-network dependencies.

Key idea

Each local draw uses only the CPT of that node and the sampled values of its parents.

Worked Prior Sample

Suppose the random draws produce:

$$O = \textit{true} \quad \text{because } u_1 < 0.10.$$

$$R = \textit{false} \quad \text{because } u_2 > 0.20.$$

The CPTs now give

$$P(C = + \mid O, \neg R) = 0.95, \quad P(L = + \mid O, \neg R) = 0.98.$$

Two more random draws might therefore produce

$$(O, R, C, L) = (\textit{true}, \textit{false}, +, +).$$

Key idea

Prior sampling never constructs the full joint table; the network itself generates samples from the joint distribution.

Prior Sampling: Benefits and Pitfalls

Benefits

- ▶ simple to implement;
- ▶ every sample is usable for unconditional queries;
- ▶ naturally exploits the network factorization;
- ▶ easy to parallelize.

Pitfalls

- ▶ does not force observed evidence to occur;
- ▶ rare events may appear almost never;
- ▶ conditional queries can waste most samples.

Key idea

Prior sampling is excellent for the model's natural distribution and poor when the query conditions on unlikely evidence.

Evidence Changes the Sampling Problem

The robot does not usually ask

$$P(O).$$

It asks

$$P(O \mid C = +, L = +).$$

One simple strategy:

1. sample complete worlds from the prior;
2. reject any world inconsistent with the observed sensor evidence;
3. estimate the posterior from the remaining worlds.

Key idea

Rejection sampling converts prior samples into conditional samples by discarding worlds that contradict the evidence.

Rejection Sampling

For query variable X and evidence e :

Algorithm

Repeat N times:

1. draw a complete prior sample s ;
2. if s disagrees with e , discard it;
3. otherwise increment the count for $X(s)$.

Normalize the retained counts.

$$\hat{P}(X \mid e) = \frac{N(X, e)}{N(e)}.$$

Key idea

The method is conceptually clean because accepted samples are distributed according to the desired conditional distribution.

Worked Rejection Example

Evidence:

$$C = +, \quad L = +.$$

Imagine 10,000 prior samples:

Sample outcome	Count
agrees with $C = +, L = +$	995
rejected	9,005
accepted and $O = \text{true}$	898
accepted and $O = \text{false}$	97

Then

$$\hat{P}(O \mid c, l) = \frac{898}{995} \approx 0.903.$$

Key idea

The estimate is close to the exact 0.902 result—but more than 90% of the simulation work was discarded.

Rare Evidence Can Destroy Rejection Sampling

If the probability of the evidence is

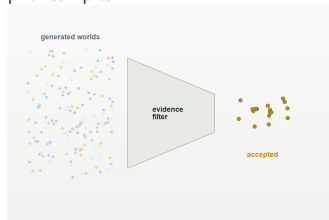
$$P(e) = 0.001,$$

then only about one sample in one thousand is expected to survive.

To obtain roughly 1,000 accepted samples, we may need around

1,000,000

prior samples.



Key idea

Rejection sampling spends computation generating exactly the worlds we already know did not happen.

Rejection Sampling: Benefits and Pitfalls

Benefits

- ▶ extremely simple;
- ▶ accepted samples have equal weight;
- ▶ easy to reason about and debug;
- ▶ good when evidence is common.

Pitfalls

- ▶ acceptance rate is approximately $P(e)$;
- ▶ rare evidence wastes almost all samples;
- ▶ high-dimensional evidence can make acceptance collapse.

Key idea

Rejection sampling is not bad because it is approximate; it is bad when the conditioning event is hard to observe by chance.

Can We Stop Throwing Evidence Away?

We already know

$$C = +, \quad L = +.$$

So likelihood weighting does not sample those variables at all.

Instead:

- ▶ lock evidence variables to their observed values;
- ▶ sample only the non-evidence variables;
- ▶ attach a weight measuring how compatible the sampled world is with the evidence.

Key idea

Likelihood weighting forces every sample to respect the evidence, then compensates statistically with a weight.

Likelihood Weighting Algorithm

Start each sample with weight

$$w = 1.$$

Traverse variables in topological order:

- ▶ if X_i is not evidence, sample it from $P(X_i \mid \text{Parents}(X_i))$;
- ▶ if $X_i = e_i$ is evidence, keep e_i fixed and update

$$w \leftarrow w P(e_i \mid \text{Parents}(E_i)).$$

Therefore

$$w = \prod_{E_i \in e} P(e_i \mid \text{Parents}(E_i)).$$

Key idea

Samples that make the observed evidence plausible contribute more to the estimate.

Worked Weight: Obstacle, No Rain

Evidence is fixed:

$$C = +, \quad L = +.$$

Suppose we sample

$$O = \textit{true}, \quad R = \textit{false}.$$

Then

$$\begin{aligned} w &= P(C = + \mid O, \neg R)P(L = + \mid O, \neg R) \\ &= (0.95)(0.98) = 0.931. \end{aligned}$$

This sample contributes weight 0.931 to the $O = \textit{true}$ bucket.

Key idea

We never reject this sample; its influence is proportional to how well it explains the observed sensors.

A Weak Explanation Gets a Small Weight

Suppose instead we sample

$$O = \textit{false}, \quad R = \textit{false}.$$

The evidence likelihood is

$$w = (0.05)(0.02) = 0.001.$$

Compare:

$$\frac{0.931}{0.001} = 931.$$

The first world contributes 931 times as much weight to the posterior estimate.

Key idea

Likelihood weighting preserves unlikely samples without pretending that they explain the evidence well.

Weighted Counts Produce the Posterior

After many samples, accumulate weights instead of raw counts.

Example:

Query value	Total accumulated weight
$O = \text{true}$	89.9
$O = \text{false}$	9.8

Normalize:

$$\hat{P}(O \mid c, l) = \frac{89.9}{89.9 + 9.8} \approx 0.902.$$

Key idea

The weights replace rejection: every sample participates, but not every sample participates equally.

Pitfall: Weight Degeneracy

For many evidence variables,

$$w = \prod_i P(e_i \mid \text{Parents}(E_i)).$$

Products of many probabilities can make most weights extremely small while a few samples dominate.
A useful diagnostic is the effective sample size

$$ESS \approx \frac{(\sum_i w_i)^2}{\sum_i w_i^2}.$$

Example

We may generate 10,000 weighted samples but obtain an ESS of only a few hundred because nearly all useful weight sits on a small subset.

Key idea

Keeping every sample is not the same as getting useful information from every sample.

Likelihood Weighting: Benefits and Pitfalls

Benefits

- ▶ never rejects evidence-inconsistent samples;
- ▶ often much better than rejection for rare evidence;
- ▶ simple extension of ancestral sampling;
- ▶ easy to parallelize.

Pitfalls

- ▶ weights can become highly unequal;
- ▶ evidence far downstream may be hard to explain;
- ▶ effective sample size can be far below sample count.

Key idea

Likelihood weighting fixes wasted rejection but can replace it with a different failure mode: concentrated weights.

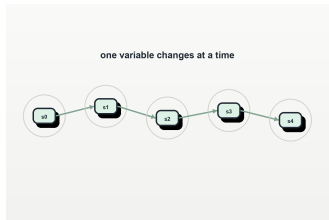
A Different Strategy: Walk Through Possible Worlds

Prior, rejection, and likelihood weighting repeatedly build a world from scratch.

Gibbs sampling instead:

1. fixes the evidence variables;
2. starts with any assignment to the remaining variables;
3. repeatedly resamples one non-evidence variable conditioned on the current values of the others.

The result is a **Markov chain** of related assignments.



Key idea

Gibbs sampling explores the posterior directly rather than hoping prior samples happen to match the evidence.

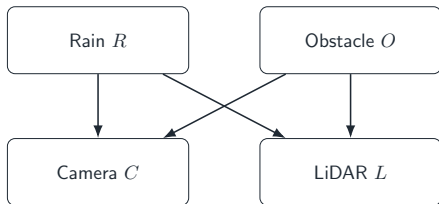
Only the Markov Blanket Is Needed

To resample a variable X , Gibbs sampling needs only its **Markov blanket**:

- ▶ parents of X ;
- ▶ children of X ;
- ▶ other parents of those children.

For obstacle O in our robot network:

$$MB(O) = \{C, L, R\}.$$



Key idea

Local graph structure makes each Gibbs update cheaper than recomputing the entire joint distribution.

Worked Gibbs Update: Resample the Obstacle

Keep evidence fixed:

$$C = +, \quad L = +.$$

Suppose the current state has $R = \text{true}$.

Score the two possible obstacle states:

$$O = \text{true} : (0.10)(0.85)(0.90) = 0.0765,$$

$$O = \text{false} : (0.90)(0.25)(0.20) = 0.045.$$

Normalize:

$$P(O = \text{true} \mid R = \text{true}, c, l) = \frac{0.0765}{0.0765 + 0.045} \approx 0.630.$$

Key idea

One Gibbs update draws a new value from the variable's conditional distribution given its current neighborhood.

Then Resample Another Hidden Variable

Suppose the chain currently has $O = \text{true}$.

For rain:

$$R = \text{true} : \quad (0.20)(0.85)(0.90) = 0.153,$$

$$R = \text{false} : \quad (0.80)(0.95)(0.98) = 0.7448.$$

Therefore

$$P(R = \text{true} \mid O = \text{true}, c, l) = \frac{0.153}{0.153 + 0.7448} \approx 0.170.$$

Alternate updates to O , R , and any other hidden variables; record their visited states.

Key idea

The chain gradually spends time in assignments in proportion to their posterior probability.

Gibbs Has Different Failure Modes

Burn-in

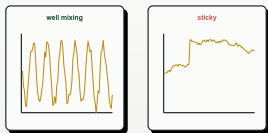
Early samples can reflect the arbitrary starting state more than the target posterior.

Autocorrelation

Consecutive samples are related, so N Gibbs samples contain less information than N independent samples.

Slow mixing

Strong dependencies can trap the chain in one region of state space for a long time.



Key idea

Gibbs avoids rejection and weight collapse, but the price is dependence between samples and uncertain convergence speed.

Same Goal, Different Failure Modes

Method	Best feature	Main failure mode
Prior sampling	simplest; all samples usable for priors	poor conditional inference for rare evidence
Rejection sampling	accepted samples are easy to interpret	acceptance collapses as $P(e)$ becomes small
Likelihood weighting	forces evidence; parallelizable	weight degeneracy / low ESS
Gibbs sampling	explores posterior directly	burn-in, autocorrelation, slow mixing

Key idea

There is no universally best sampler; each method moves the computational difficulty somewhere different.

Takeaways: Approximation Is an Engineering Decision

- ▶ Exact inference gives a deterministic answer but can become computationally expensive.
- ▶ Prior sampling estimates probabilities from worlds generated by the Bayesian network.
- ▶ Rejection sampling conditions by discarding samples that disagree with evidence.
- ▶ Likelihood weighting fixes evidence and assigns each sample an evidence likelihood weight.
- ▶ Gibbs sampling forms a Markov chain by repeatedly resampling non-evidence variables from local conditional distributions.
- ▶ Sample count alone is not enough: acceptance rate, weight concentration, and mixing determine how much information the computation actually produced.

Key idea

Approximate inference is not “less rigorous” inference: it explicitly trades exactness for runtime and measures the consequences of that trade.